Knows the Video, Not the Moment: Temporal Localisation Is the Missing Capability of Streaming Video Anomaly Detection

Anonymous CVPR submission

Abstract

*Streaming video anomaly detection (VAD) has to decide, every two seconds and from eight frames, whether something abnormal is happening now. Small decision-tuned vision-language models do this in one forward pass at a fifth of real time, and a short fine-tune lifts a 2.2B model from 82.7 to 87.6 frame-level AUC on UCF-Crime. We show that this number measures the wrong thing. Every model we evaluate, from 0.8B to 27B, zero-shot or fine-tuned, tells anomalous videos from normal ones at 96 to 97 AUC and ranks the moments inside them at 72 to 74: the windows of an anomalous video outside its annotated events score 0.43 on average, the normal videos' windows 0.07, and the gap does not shrink with distance from the event. The model knows the video, not the moment. Fine-tuning makes this worse while the task metric improves: the standard hard-label fine-tune flattens the inside-outside contrast (+0.25 to +0.09), triples the calibration error, drops MVBench by 10 points and transfer by 4. We introduce **jev-hard**, which stratifies the events of UCF-Crime and XD-Violence by a model-free size proxy and scores each tier three ways: window AP, event recall at a false-alarm rate, and a threshold-free event AP. The three disagree, and the disagreement is informative: under window AP small events are a wall that scale does not move and nine of twelve classes sit at chance; under the event metrics the small tier is no harder than the others and a 9B model catches 1.3 to 1.9 times the events of the 2.2B one at the same false-alarm rate. We then measure what each cheap intervention moves. A multi-task mix with a KL anchor keeps the detection gain without the narrowing and transfers 7 points above zero-shot; weight interpolation retains everything but transfer; conditional foveation lifts the small tier by magnification, with a motion-area gate that protects scene-wide events. None of them improves localisation inside the video. Per-window scores, the benchmark and the protocol are released.*

1. Introduction

Online video anomaly detection (VAD) answers one question every few seconds, from past frames only: *is something abnormal happening now?* Recent methods answer it by generating and scoring text [14, 28, 30, 32], at seconds per decision. *Decision models* [3, 23, 36] answer with a probability over fixed options in one forward pass. A 2.2B one scores a 2 s window of eight frames in 354 ms at 82.7 frame-level AUC on UCF-Crime zero-shot; one epoch of LoRA on 20 k windows lifts it to 87.6, within a point of the trained online state of the art [2]. This paper asks what that number measures, and finds that it is mostly not the thing a streaming detector is for.

The score lives at the video level. Taking each test video's maximum window score, every model we evaluate, from 0.8B to 27B, zero-shot or fine-tuned, decision-trained or plain, separates anomalous videos from normal ones at 95.5 to 97.3 AUC. Inside the anomalous videos the same models rank the annotated moments above the rest at only 72 to 74 AUC. The windows of normal videos average 0.05 to 0.08; the unlabelled windows of anomalous videos average 0.43, and this does not decay with distance from the nearest event (0.49 one window away, 0.42 eleven or more away). The models recognise a video in which a crime happens, and carry that recognition across the whole video; they do not localise the crime in time (Sec. 4). Everything that follows is a consequence.

Fine-tuning narrows the model and flattens the contrast. The task's own metric cannot see what a fine-tune takes away. On eleven gates (Tab. 2), the standard hard-label fine-tune triples the calibration error on the model's own benchmark, drops MVBench by 9.6 points, raises false alarms on anomaly-free video, loses 3.6 AUC on unseen scenes and 4 points of transfer; and it lowers the inside-minus-outside score contrast of anomalous videos from +0.25 to +0.09 while raising AUC by five. The remedy for the narrowing needs no new annotation, a multi-task mix under a KL anchor to the frozen release [11], and the honest baseline, weight interpolation [24], retains almost as much; neither restores the contrast.

The metric decides what "small" means. We stratify the 156 annotated events of UCF-Crime, and the 1,238 of

XD-Violence, by a model-free measure of how much of the frame moves (Sec. 3.3) and score each tier three ways. Under window AP the small tier is a wall: 16 to 29 AP for every model, nine of twelve classes at chance, two points from 4.5× the parameters. Under event recall at a fixed false-alarm rate, or a threshold-free event AP, the small tier is no harder than the others, half of its events are noticed, and a 9B model catches 1.3 to 1.9 times the events of the 2.2B one. The window metric is measuring localisation, which no model has; the event metrics measure noticing, which scale buys.

What the cheap interventions move. Conditional foveation, four of eight frames replaced by crops on the motion peak when little moves, trained in on the non-narrowing recipe, lifts the small tier by 4.8 AP over three seeds; a gate-less full-frame up-scaling gets most of that and loses 12 points on scene-wide events, so the gain is magnification and the gate is what makes it affordable. Hard negatives trade the medium tier for the small one along a smooth dial. Inference-time zoom, the model's own gate and its own saliency, DPO, joint training on both datasets, and video clips as replay all lose or break a gate. None of the interventions, nor scale, moves the within-video AUC by more than two points.

Contributions.

- **A diagnosis**: streaming VAD scores separate videos, not moments; the missing capability is temporal localisation inside anomalous video, and fine-tuning for the task erodes it while the task metric rises (Sec. 4).
- **jev-hard**, an event-size stratification of two datasets scored as window AP, event recall at a false-alarm rate and threshold-free event AP, reported with retention gates, whose disagreements locate what each metric measures (Secs. 3.3 and 6.3).
- **A measurement of narrowing and of every cheap repair**: the multi-task recipe, interpolation, foveation with five controls, and six negative results, each with seeds or bootstrap intervals (Secs. 6.2 and 6.4).

All comparisons concern the policies and the 0.8B to 27B backbones we evaluate; X marks a run not made.

2. Related Work

Online VAD. Weakly supervised detectors score snippets from video-level labels [21, 22]; REWARD [7] makes this real-time and ReactVAU [2] pairs a light Yes/No detector with a heavier reasoner. Language-based methods caption a window and score the caption [30, 32]; MoniTor [28] and ESOM [14] make them streaming, StreamForest [33] keeps an event memory. We take their questions, causal windows and frame-level metrics, and study adaptation of a model that answers in one pass. Methods without per-window scores cannot be placed on jev-hard; their absence implies no ordering.

Decision models. StartLux-Decision [36], Decision 3.0 [23] and Clef-Flash [3] are open VLMs post-trained to return a probability for every option of a typed question, from option-letter logits, an answer-code head or a joint head over question and option spans. We use them as given; the typed interface makes calibration measurable, it does not make it hold.

Forgetting and its remedies. Our KL anchor to the frozen release is distillation in the sense of LwF [11] and our replay is rehearsal [19]; the contribution is the measurement along eleven gates, the recipe without new annotation, and its replication across read-out families. WiSE-FT [24], DARE [31] and TIES [27] are the zero-training alternatives; we report the full WiSE-FT curve as a baseline and interpolate two of our own adapters to trace a trade-off.

Resolution, zoom and glimpses. Mini-Gemini [10], LLaVA-HR [15] and Monkey [12] add resolution architecturally and spend it on every input; ViCrop [35] crops where the model's attention looks, at inference; glimpse models [16] learn where to look. Our foveation is at the data level, decided by a model-free motion signal, trained and run with the same transformation under causal streaming constraints; we do not claim a fixed token count as our own. Top-k instance selection [21, 22] is the ancestor of our hard-negative mining; focal loss [13] is its untested alternative.

3. Task, Models and the jev-hard Protocol

3.1. Streaming protocol

A stream is processed in windows of $W = 2\,\mathrm{s}$ with stride W. Window t holds the frames of $[2t, 2t + 2)\,\mathrm{s}$, sampled uniformly to $F = 8$ frames. At the end of window t the model receives the eight frames and a question and returns a probability p_t that the window is abnormal. Every frame of the window takes p_t as its score, which gives the frame-level AUC and AP that the VAD literature reports [21, 25]. No future frame is read and nothing is smoothed over the video. Latency is the wall-clock time from the last frame of the window to p_t, one window at a time on one GPU; we report its 95th percentile and the real-time factor $\mathrm{RTF} = \mathrm{P95}/W$. Edge windows are only partly abnormal and UCF-Crime's annotations are inclusive at both ends; we keep the standard frame-level labels for comparability and use the partial coverage as a training target (Sec. 5.1).



Figure 1. **jev-hard.** An annotated event is probed by five windows; the TGM hot area of each probe sizes it; the median over probes places it in a tier. A tier's score takes the tier's windows as positives and keeps every negative window of the test set, so tier AP is comparable to the all-video AP. Submissions co-report the generalisation suite of Sec. 6.2.

3.2. Decision models and read-outs

A decision record is a *state* (the eight frames and a one-line description), a *question* and its *options*; a decision model returns $p(o \mid \text{state}, \text{question})$ for every option in one forward pass. The models we adapt (supplementary Tab. 5) read this out through a linear head over answer codes (Decision 3.0 [23]: d3-nano 2.2B, d3-flash 8.4B, d3 27B), a small transformer head over question and option spans (CleFlash [3], 9B), or the first-token logits of the options (the plain Qwen3.5-2B and 9B instruction models, our controls). All share the Qwen3.5 hybrid backbone. The question follows MoniTor [28] (**V1**): the dataset's anomaly classes named with one-line definitions, asked as one Yes/No question; the score is $p(\text{Yes})$. A second question asks *what is happening* with the 13 UCF-Crime classes and *Normal* as options (**14-class**); it is never used for scoring, only to test whether a fine-tune can add an ability. Full prompts are in the supplementary material.

3.3. jev-hard: events by the size of what moves

Aggregate AUC and AP hide which events a model misses. We stratify the annotated test events by a model-free proxy of their spatial extent (Fig. 1). For every event, five probe windows are spread over its annotated span; in each, the temporal gradient magnitude (TGM) of the eight frames, the absolute frame difference summed over the window, is thresholded at half its peak, and the *hot area* is the fraction of the frame above threshold. The event's size is the median over its probes. Terciles over the dataset's events give three tiers: on UCF-Crime, *small* below 2.51% of the frame, *medium* to 4.46%, *large* above, 52 events each; on XD-Violence the 1,238 events are split at their own terciles (7.0% and 16.8%). The same measurement on 591 annotated events of the UCF-Crime *training* split gives 2.31% and 4.33%, so the cut-points are a property of the data, not of the test set, and the 4.5% used by the foveation gate of Sec. 5.3 is the training-set value rounded. The frames the model sees are never cropped by the measurement; TGM only sizes the event.

A tier is scored three ways, and Tab. 1 adds the within-video AUC that measures localisation directly. **Window AP**: the tier's windows are the positives and *every* negative window of the test set is kept, so the three tier scores share the negatives of the all-video score; windows of other tiers' events are dropped. **Event recall at a false-alarm rate**: an alarm is a window whose score reaches a threshold τ; τ is set per model so that the normal test videos (6.0 h on UCF-Crime, 14.5 h on XD-Violence) raise k alarms per hour, and an event counts as caught if any of its windows alarms. **Event AP**, threshold-free: each event scored by its maximum window against every window of the normal videos. The first asks whether the moment is ranked above all other footage, the other two whether the event is noticed at a cost the operator accepts; Sec. 6.3 shows that they disagree. All scores are recomputable offline from per-window outputs. A third rule: a tier score is reported together with the generalisation gates of Sec. 6.2, so that a method cannot win the hard tier by narrowing the model. Motion area is a proxy that confounds extent with duration, class and camera motion: the small tier leans to long, distant, slow events (median 11 s), the large tier to short scene-wide ones (median 5 s), and a long event contributes more windows to the window metric. A human check of the small tier is in progress.

4. Where the Score Lives

Figure 2 shows what a window score looks like over a video, and Fig. 3 and Tab. 1 quantify it over the 139 anomalous test videos that contain both labelled and unlabelled windows. Three numbers describe every model in the study. **Video-level AUC**, each video scored by its maximum window, is 95.5 to 97.3: anomalous videos are recognised. **Within-video AUC**, labelled against unlabelled windows inside the anomalous videos, is 68.8 to 74.4: the moment is barely ranked. The **mean score of unlabelled windows** inside anomalous videos is 0.33 zero-shot and 0.41 to 0.47 after any fine-tune, against 0.05 to 0.08 for the windows of normal videos. The all-video window AP that the field reports (41.9 zero-shot, 47.4 after the recipe) is almost entirely the second number: pooled over the anomalous videos alone it is 49.

Why the unlabelled windows score high. Four explanations were tested. *Loose annotation boundaries* would make the score decay with distance from the event; it decays only from 0.49 to 0.42 between one window and eleven or more windows away for the recipe, and from 0.45 to 0.42 for the hard-label fine-tune, against 0.45 to 0.30 zero-shot (Tab. 1, last column): fine-tuning removes what little decay the release had. *A scene prior* would make a video's unlabelled score track its labelled score and its zero-shot score; across videos the two correlate at 0.61 to 0.80, and every fine-tuned model's unlabelled scores correlate at 0.62 to 0.73 with the zero-shot model's: the prior is in the release and the fine-tunes inherit it. *Motion rather than anomaly* is

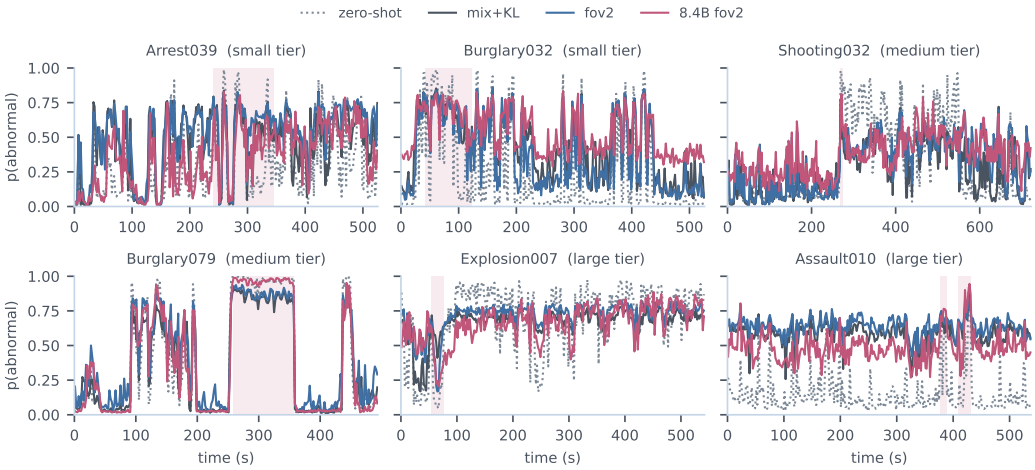


Figure 2. **Six anomalous test videos, two per tier.** Window scores of four models against time; shaded, the annotated events. In most videos the score is high throughout, and the event is a bump or nothing at all (Explosion007); in a few (Burglary079) it is localised. The fine-tuned models raise the whole trace, the zero-shot model is lower and noisier.

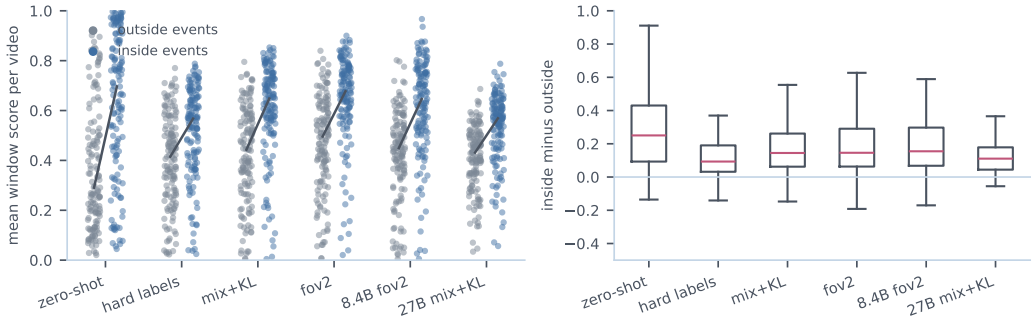


Figure 3. **Inside against outside, per anomalous video (139 videos with both kinds of window).** Left: each video's mean score inside its events (blue) and outside them (grey), medians joined. Right: the inside-minus-outside difference. Fine-tuning raises both means and compresses the difference; scale does not widen it.

d3-nano unless stated	video AUC	within-video AUC	median in − out	unlab. $\geq .5$	dist 1 / 11+
zero-shot	95.5	71.9	+0.25	28%	0.45 / 0.30
hard labels	96.7	68.8	+0.09	39%	0.45 / 0.42
mix + KL	96.1	73.3	+0.14	48%	0.49 / 0.42
WiSE-FT α=.5	97.0	73.6	+0.12	46%	0.53 / 0.46
fov2	96.4	74.0	+0.15	54%	0.54 / 0.47
fovdial α=.25	96.9	74.4	+0.15	46%	0.51 / 0.43
d3-flash (8.4B), fov2	97.2	72.7	+0.15	38%	0.48 / 0.41
Qwen3.5-9B, mix + KL	97.2	73.8	+0.18	31%	0.45 / 0.36
d3 (27B), mix + KL	97.0	70.8	+0.11	36%	0.45 / 0.41

Table 1. **Video level against window level (UCF-Crime).** Video AUC: anomalous against normal videos by maximum window score. Within-video AUC: labelled against unlabelled windows inside anomalous videos, pooled. Median over videos of the inside-minus-outside mean score; share of unlabelled windows inside anomalous videos scoring ≥ 0.5; mean score of unlabelled windows one window from an event and eleven or more away.

tested in supplementary Tab. 11 by TGM area; the unlabelled windows of anomalous videos score the same across motion quartiles. *Training labels*: the training windows outside events in anomalous videos are labelled 0 under every recipe, and the hard-label and mix + KL models score them identically at test (0.42); the label is there, the model does not learn it. The reading is that the model has learnt that a crime video is a crime video, which is nearly all of what the frame-level metric rewards.

What this means for the metric. A streaming detector raises an alarm at a moment and is judged by whether the moment was right and how often it was wrong. Frame-level AP, computed over all windows of a test set in which anomalous videos are a third, rewards the video prior: a model that scores every window of an anomalous video at 0.9 and every window of a normal video at 0.1 reaches a high AP without localising anything. The event-level scores of Sec. 6.3 remove the prior by counting events and false

alarms; the within-video AUC of Tab. 1 measures localisation directly. We propose that streaming VAD report both, and we use them for the rest of the paper.

5. Adapting Without Narrowing, Then Looking Closer

5.1. Window records from annotated training videos

The training splits of UCF-Crime (1,610 videos) and XD-Violence (3,954) carry one label per video. Holmes-VAU [34] added start and end times for every anomaly event in 735 anomalous UCF-Crime and 2,046 XD-Violence training videos. From these we cut the same 2 s windows of eight frames as at inference and label each window by its *coverage* $y \in [0, 1]$, the share of the window inside an event (supplementary Fig. 6); the window's class is the video's category, so the 14-class question is labelled at no cost. Negatives come from normal videos and the normal stretches of anomalous videos. A run samples 20 k windows, balanced 1:1, and holds out 10% of the training *videos* for a temperature. No test video or annotation is read at any point.

5.2. The fine-tune and what it is compared with

Every arm trains the same thing: vision tower frozen, a rank-64 LoRA [6] on every language-model projection, the read-out head where the release has one, the release's own prompt mask, one epoch of AdamW (peak 1.5×10^{-4}, 10^{-4} at 9B and above), a temperature fitted on the held-out videos, the adapter merged. Nothing changes at inference. The arms differ in data and loss:

- **Hard labels** (today's recipe): the V1 question only, cross-entropy on $y \geq 0.5$.
- **Mix**: 50% V1 records with the soft coverage target, 25% the same windows asked the 14-class question, 25% replay of the release's own text decision benchmark [18] (choice, Yes/No and score items, their own labels), all under the soft cross-entropy over the model's own option read-out.
- **Mix + KL**: the mix, plus a second frozen forward of the release on every record and a penalty on drift at the answer position,

$$\mathcal{L} = \mathrm{CE}_{\mathrm{mix}} + \lambda\, \mathrm{KL}\big(p_{\mathrm{release}} \,\|\, p_\theta\big), \qquad \lambda = 0.5, \tag{1}$$

  which is distillation from the model's own past in the sense of LwF [11] (supplementary Fig. 7).
- **DPO**: the mix data as preference pairs, correct option chosen against a wrong one, with the implicit KL of DPO as the only drift control.
- **WiSE-FT** [24]: no training; the hard-label adapter scaled by $\alpha \in \{0.25, 0.5, 0.75\}$, which interpolates release and fine-tuned weights.

5.3. Conditional foveation

The small tier is a resolution problem: at the tower's input size a distant event is a few pixels. Cropping the moving region, up-scaling it and asking again lifts individual events (supplementary Sec. 18), but on the full test set no inference-time policy beats the un-zoomed model, whether the crop is taken on every window or only on uncertain ones: the model has never seen a crop. We therefore make the crop part of the input format and train with it (supplementary Fig. 11).

Where and when. The TGM map of the window (Sec. 3.3) is computed once; its peak is the crop centre, and the crop is a third of the frame around it, up-scaled 2×. If the moving area covers at least 4.5% of the frame (the medium–large boundary, from the training split) the event is scene-wide and the model sees the eight full frames; otherwise frames 1, 3, 5 and 7 are replaced by the crop and 0, 2, 4, 6 stay full, so the global view is never lost. Frame budget, token count and model latency are unchanged; the rule is applied identically in training and at inference (**fov2**). Its controls: always alternate (**fov1**); the whole frame up-scaled on the same slots, no gate (**full-frame**); a centre crop (**fovcen**); the model's own uncertainty at window t as the trigger for $t{+}1$ (**fov3**); and its gradient saliency as the position (**fov4**), one backward pass per window.

Hard negatives and the dial. A crop magnifies crime-like normal behaviour as much as small events. **fov2h** adds 2,500 windows from normal training videos that the mix + KL model itself scores highest (at most 40 per video), labelled *No*. fov2 and fov2h share initialisation and data otherwise, so their adapters interpolate element-wise, $\theta_\alpha = (1-\alpha)\,\theta_{\mathrm{fov2}} + \alpha\,\theta_{\mathrm{fov2h}}$ (**fovdial**); the curve over α is a post-hoc analysis, not a selected operating point.

6. Experiments

6.1. Setup

Datasets and metrics. UCF-Crime [21]: 290 test videos, 13 anomaly classes, 18,390 windows of 2 s; XD-Violence [25]: 800 test videos, six classes, 48,408 windows, AP as its main metric. Frame-level AUC and AP as defined in Sec. 3.1; tier scores as in Sec. 3.3. Every number comes from one scoring pipeline over released per-window outputs, except rows marked †, scored once by an earlier pipeline (Tab. 2).

Generalisation suite. Eleven gates measure what a fine-tune of d3-nano does outside its task: 14-class window accuracy; UCF101 [20] top-1 as a typed choice; accuracy and 10-bin ECE on the 22 k items of the release's own decision

benchmark [18]; MVBench [9] and Video-MME short [4]; false alarms, the Yes-rate of V1 at $\tau = 0.5$ on UCF101, which has no anomaly; frame AUC on UBnormal [1], unseen scenes; and transfer to XD-Violence and back.

Hardware and seeds. Latencies are measured one window at a time on one NVIDIA RTX PRO 6000 with reference DeltaNet kernels (fast kernels give 0.6 to 0.75 of the tabled values). The mix + KL arm and fov2 have three seeds on d3-nano; every other fine-tune is one seed, and differences within 2 AP are not interpreted.

6.2. Adaptation Narrows the Model, and Need Not

Supplementary Tab. 7 places the models among published online methods. Table 2 is the central measurement (supplementary Fig. 9 draws it). The hard-label fine-tune lifts d3-nano from 82.7 to 87.6 AUC on its task and, on the same model, triples the calibration error on the release's own benchmark (0.070 to 0.221), drops MVBench by 9.6 and Video-MME by 7.9 points, raises false alarms on anomaly-free video from 4.3% to 6.8%, loses 3.6 AUC on unseen scenes and about 4 points of transfer in each direction; its AP on the single pipeline falls (41.9 to 35.6), and Tab. 12 shows where: on the small tier. On the localisation axis of Tab. 1 the hard-label fine-tune compresses the inside-minus-outside contrast from +0.25 to +0.09 and lowers the within-video AUC (71.9 to 68.8) while raising the all-video AUC by five: the task metric rewards the video prior it strengthens. The mix alone repairs calibration and most of the video understanding but loses the AP (34.4); the KL anchor restores it (47.4) with ECE 0.068, false alarms 4.9% and MVBench 53.9, two thirds of the way back, with 3.6 MVBench, 3.8 Video-MME and 1.8 UBnormal points still open. DPO reaches the same detection (45.9 AP) but its ECE is 0.159 and its temperature saturates: a preference loss moves the ratio between option logits and says nothing about their scale.

WiSE-FT is the result a reviewer should ask for, and it is strong: with no training, $\alpha = 0.5$ reaches 89.1 AUC and 47.8 AP, above mix + KL on detection, and $\alpha = 0.25$ has the best ECE (0.060), MVBench (56.0) and Video-MME (63.0) of any adapted model. What it cannot do is add an ability (the hard-label adapter never saw the 14-class question; no α produces it, 86.8% against 90.0%) or a new input format (Sec. 6.4), and it transfers worse (78.2 against 86.7 AP on XD-Violence); that is why the mix + KL recipe is the base of the foveated models.

Supplementary Tab. 6 replicates the recipe: on the plain Qwen3.5-2B the mix + KL arm lifts AP by 12 points where the hard-label fine-tune lifts none; at 9B all three read-out families land between 88.3 and 89.1 AUC and 48 and 49 AP, the plain backbone not behind; the 27B model reaches 88.0 / 46.6, the level of the 2.2B one (88.4 $\pm$ 0.5 / 45.2 $\pm$ 2.1) at twelve times the cost. The transfer gate turns around entirely: the mix + KL adapter trained on UCF-Crime scores 86.7 AP on XD-Violence, 11 points above the hard-label adapter (75.7) and 7 above zero-shot (79.7) on the same 48,408 windows (the earlier pipeline ranks them the same way), and the XD-trained adapter comes back to UCF-Crime at 84.4 AUC against 78.7 for the hard-label one; the recipe learns the anomaly question rather than the dataset. Replacing a quarter of the text replay by UCF101 video clips, the obvious attempt at the remaining video-QA gap, changes nothing (MVBench 53.4, Video-MME 59.7): the loss is not repaired by more video of that kind. Training one model on both datasets at once (10 k windows each, the same mix) keeps UCF-Crime (88.3 / 46.8) and XD-Violence (96.8 / 91.8) near their single-dataset floors but loses 6 AUC on unseen scenes (UBnormal 62.9 against 68.9), a negative result we report as such.

6.3. Events, Not Windows

Under window AP (supplementary Tab. 12) every model reads the same way: small below medium below large, a 13-point AUC spread zero-shot, and a small-tier AP between 16 and 28 that neither size nor recipe moves; the 27B model trained with the recipe reaches 24.3 there, the level of the 2.2B one (22.6 $\pm$ 0.2). Per class it is starker (supplementary Tab. 9): with every negative window as denominator, chance is 0.6% AP and nine of the twelve classes with small events sit between 0.1 and 5 for every model; the tier's 27 AP is carried by Stealing (65 windows, 46 to 82 AP across models) and Explosion.

The events are caught. At the event level the wall is not there. With $\tau = 0.5$, 41 of the 52 small events have a window that fires zero-shot and 50 after foveation; at $\tau = 0.3$ every one does. Table 3 fixes the false-alarm rate instead: at one alarm per hour of normal footage the 2.2B models catch 14 to 29 small events, the 9B ones 25 to 35, the 27B one 28 to 29 zero-shot, and the small tier is no worse than the other two. The thresholds are fragile (chosen on half of the normal videos they give 0.7 to 11 alarms per hour on the other half, 11 for the zero-shot model), so the table also gives the threshold-free event AP, which agrees. With Sec. 4 in mind the two readings are consistent: window AP inside anomalous videos measures localisation, which no model has, and the event metrics measure noticing. Two window-level conclusions therefore need their metric named: *scale does not move the small tier* holds for window AP, while at the event level a 9B model with foveation catches 1.3 to 1.9 times the events of the 2.2B one and a 27B model catches more zero-shot than the 2.2B one does fine-tuned; and *the small tier is the hardest* is about ranking windows, not noticing events.

			Single task (V1 only)		Multi-task mix				Weight interpolation		
Gate	Metric	Zero-shot	hard labels	DPO	mix	mix, hard	mix+KL	mix+KL, hard	WiSE .25	WiSE .5	WiSE .75
1. Detection	UCF-290 frame AUC	82.7	87.6	87.8	87.1	X	87.9	X	88.3	89.1	**89.2**
	UCF-290 frame AP	41.9	35.6	45.9	34.4	X	47.4	X	46.3	**47.8**	47.0
2. New ability	UCF-290 14-class acc.†	88.5	–	89.8	89.4	X	**90.0**	X	X	86.8	X
3. Nothing lost	jev-bench ECE†	0.070	0.221	0.159	0.063	X	0.068	X	**0.060**	0.102	0.161
	MVBench†	**57.3**	47.7	50.6	52.5	X	53.9	X	56.0	54.2	51.8
	Video-MME short†	**63.9**	56.0	58.9	58.9	X	60.6	X	63.0	60.3	X
	False alarms on UCF101 (%)†	**4.3**	6.8	6.7	7.4	X	4.9	X	X	1.7	X
	UBnormal frame AUC†	70.7	67.1	69.3	68.8	X	68.9	X	**72.8**	70.9	X
	UCF→XD frame AP	79.7	75.7	X	X	X	**86.7**	X	78.2	78.7	X

Table 2. **Fine-tuning narrows the model; the multi-task mix with a KL anchor does not (d3-nano, 2.2B).** Every column is one LoRA fine-tune of the same release on the same 20 k UCF-Crime training windows, except WiSE-FT, which interpolates the release and the hard-label adapter at α with no training. Gate 1: frame-level AUC and AP on the 290 UCF-Crime test videos. Gate 2: window accuracy on the 14-class question the single-task recipes never saw. Gate 3: abilities outside the task, in %, except ECE (window-level, 10 bins); false alarms are the Yes-rate at $\tau = 0.5$ of the V1 question on UCF101, which contains no anomaly; UCF101 top-1 and jev-bench accuracy, which no arm moves by more than 2 points except the mix (UCF101 -4.7), are in the supplementary material. All AUC / AP are from one scoring pipeline; † rows are measured once, on the deck pipeline. Bold: best per row. X: not finished at the time of writing.

		caught of 52 at 1 FA/h			event AP		
UCF-Crime	τ	small	med.	large	small	med.	large
d3-nano, zero-shot	0.98	14	23	24	52.5	59.5	66.5
d3-nano, hard labels	0.70	17	16	17	59.6	59.3	57.4
d3-nano, mix + KL	0.78	24	20	21	62.2	59.9	59.6
d3-nano, WiSE-FT α=.5	0.80	22	22	21	61.7	63.0	61.5
d3-nano, full frame up-scaled	0.81	23	22	20	61.3	58.9	57.6
d3-nano, fov2	0.80	28	21	25	67.5	61.5	58.3
d3-nano, fovdial α=.25	0.75	29	24	26	72.1	66.0	64.8
d3-flash (8.4B), mix + KL	0.70	25	35	33	73.7	80.0	81.4
d3-flash, fov2	0.72	35	**39**	36	82.6	81.8	78.5
Qwen3.5-9B, mix + KL	0.74	29	31	32	77.2	82.1	80.2
Qwen3.5-9B, fov2	0.72	**36**	36	**38**	**85.4**	**82.9**	**81.7**
d3 (27B), zero-shot	0.56	29	29	29	69.1	70.5	72.0
d3 (27B), mix + KL	0.61	28	31	34	71.4	78.3	79.3

Table 3. **Events, not windows.** Left: events caught when τ is set per model so that the 6.0 h of normal UCF-Crime test video raise six window alarms. Right, threshold-free: AP of each event's maximum window score against every window of the normal videos. Both reverse the ordering by size of the window AP (supplementary Tab. 12), and the small tier is not the hardest. Supplementary Fig. 10 gives the curves over the alarm rate.

d3-nano, mix + KL base	small	medium	large	all 290
no zoom (the base, 3 seeds)	85.2 / 22.6 ±0.2	91.3 / 34.8 ±3.2	91.5 / 21.1 ±6.9	88.4 / 45.2 ±2.1
full frame up-scaled, no gate, no localisation	85.8 / 26.2	91.5 / 39.1	90.9 / 16.7	88.7 / 46.5
fov1, always alternate	85.3 / 25.6	91.2 / 36.2	91.0 / 18.9	88.4 / 46.0
fov2, motion-area rule	85.7 / 27.1	91.4 / 36.4	91.4 / 24.0	88.7 / **47.8**
fov2, 3 seeds	86.1 / 27.4 ±0.4	91.2 / 33.5 ±2.6	90.4 / 18.4 ±4.9	88.6 / 45.9 ±1.6
fovcen, centre crop	85.1 / 24.7	90.7 / 35.5	91.5 / 27.6	88.2 / 46.8
fov2h, + hard negatives	86.0 / **29.2**	89.7 / 29.6	91.3 / 20.8	88.3 / 45.8
d3-flash (8.4B), mix + KL / + fov2, AP	24.2 / 28.4	37.0 / 37.2	31.5 / 26.5	48.2 / 49.4
Qwen3.5-9B, mix + KL / + fov2, AP	27.3 / 29.9	38.8 / 38.9	24.6 / 26.1	49.6 / 50.9
fovdial $\alpha = 0.25$	86.1 / 28.5	91.4 / 35.2	91.9 / 24.0	**89.0 / 48.2**

Table 4. **The foveation ladder (d3-nano, UCF-Crime, frame AUC / AP).** All rows are fine-tuned with mix + KL on the same windows and differ only in the input transformation; the first row has none. UBnormal transfer is in Tab. 13. The base and fov2 rows marked 3 seeds give the mean with the AP standard deviation; the seed spread lives in the large tier, the small tier is stable to 0.4 AP. Other rows are one seed. The last two rows fuse the per-window scores of two models.

6.4. What Moves Which Metric: Foveation and Scale

Table 4 isolates the mechanism one control at a time (full ladder in supplementary Tab. 14). **Inference-time zoom fails** on every policy tried (supplementary Tab. 15). **Trained in, the crop works**: fov1 lifts the small tier by 2.9 AP and shows the cost of always cropping, ten points on the large tier. **The motion-area rule is the right *when***: fov2 keeps the small-tier gain (27.4 $\pm$ 0.4 over three seeds against 22.6 $\pm$ 0.2) with the large tier inside noise; a paired bootstrap over the 290 videos puts the small-tier gain of each seed pair between +4.4 and +5.2 AP with every interval above zero; fov3, the model's own uncertainty as the trigger, does no better. **Most of the small-tier gain is magnification, the gate is what makes it affordable**: the control that up-scales the whole frame on the same four slots, with no gate and no localisation, reaches 26.2 on the small tier (fov2: 27.1, difference inside the interval) and 16.7 on the large tier, 12 points below the base, because scene-wide events lose half their full views; fov2 keeps the small-tier gain and the large tier. A centre crop loses 2.4 small-tier AP against the motion peak, and the fine-tuned model's gradient saliency loses 1.1 AUC and 1.5 AP to it on the screening subset at three times the inference cost. The gate is not tuned: moved to 3% or 6% at inference it changes every tier by at most 0.4 AP. At 2B, a rule beats the model on both decisions. **Foveation scales**: on d3-flash (8.4B) it adds 4.2 small-tier AP (24.2 to 28.4) and on the plain Qwen3.5-9B 2.6 (27.3 to 29.9, $[-0.7, +5.3]$), and the 9B foveated models give the best all-video AP (49.4 and 50.9) and the best

event recall of the study (Tab. 3). **Foveation does not narrow**: on unseen scenes fov2 scores 72.7 ± 2.4 over three seeds, above zero-shot (70.7) and above the base (68.9), and false alarms stay at the base's level. On the plain Qwen3.5-2B, fov2 adds 0.6 AUC at unchanged AP (88.8 / 45.6).

Interpolation reaches the same point. WiSE-FT at $\alpha = 0.5$, with no training, scores 27.1 on the small tier too (supplementary Tab. 12), and the bootstrap finds no difference between it and fov2 on any tier: interpolating the hard-label adapter half-way back to the release recovers the small events it lost (16.0) as well as foveation does. They add (per-window maximum: 89.4 / 49.1, two passes; supplementary Tab. 14).

On XD-Violence. Supplementary Tab. 10 repeats the study on XD-Violence (2,645 small-tier windows). Trained on XD-Violence, the hard-label fine-tune scores 52.3 small-tier AP and the mix + KL recipe 61.3 (+9 [+7.2, +11.1]), so there the narrowing fix alone recovers the small events; foveation adds nothing (+0.7 [−1.3, +2.6]) because XD's small tercile starts at 7% of the frame, above the gate, so the crop rarely fires, as the rule intends. Interpolation reaches 58.1. Transferred from UCF-Crime without XD training, the three adapters land at 86.6 to 86.7 AP, and there the foveated ones do win the small tier (+4.3 [+0.9, +7.3] for fov2, +6.7 [+3.6, +9.5] for the dial).

The dial (supplementary Fig. 8). Hard negatives under the crop win the small tier, 29.2 AP, the best of any model in the paper including the 9B ones, and lose the medium tier, 36.4 to 29.6: crime-like normal behaviour and medium-sized real crime overlap, and suppressing the first suppresses the second. The interpolation between the two adapters is single-peaked in all-video AP, 47.8 at $\alpha = 0$, 48.2 at 0.25, 47.7 at 0.5, 46.9 at 0.75, 45.8 at 1: a light dose of hard negatives regularises and a full dose over-suppresses, and the optimum lies inside the dial. The $\alpha = 0.25$ point also transfers best, 73.5 on UBnormal. We report the curve rather than select a point on the test set; the 0.4 AP between its peak and fov2 is within seed noise.

Latency. The foveated models have the frame count and token count of the base, so their model latency is the base's. Measured end to end, preprocessing included, on one quiet GPU with reference kernels over the same 595 windows twice each, the base runs at 741 and 773 ms P95 and fov2 at 749 and 797 ms: the crop's frame difference and resize cost less than the run-to-run spread. The tabled 354 ms (RTF 0.18) is the same model on a dedicated GPU.

6.5. Generalisation

Supplementary Tab. 13 is the second axis of the protocol. The zoom line improves transfer to unseen scenes where the hard-label fine-tune loses it, carries the recipe's cross-dataset transfer unchanged in both directions, and keeps MVBench at a cost of about one Video-MME point. The video-understanding gap of the base (3.6 MVBench, 3.8 Video-MME points) is the one gate no training variant closes: video clips as replay do not move it, and joint training on both datasets loses the unseen-scene gate outright. Interpolation at $\alpha = 0.25$ is the retention point of the study, within 1.3 MVBench and 0.9 Video-MME points of zero-shot, and it pays on the two axes the recipe owns: transfer (78.2 against 86.7) and the new question.

7. Limitations

The proxy. Motion area confounds extent with duration, class and camera motion, and the tiers inherit UCF-Crime's class imbalance, hence the per-class and event-level scores. The gate and the cut-point share the TGM signal; the cut-point is reproduced on the training split (4.33% against 4.46%) and the gate is insensitive to ±1.5 points. A human check of the 52 small-tier events is in progress; until then "small" means "little moves". **Seeds and selection.** The base and fov2 have three seeds on UCF-Crime and UBnormal; everything else is one measurement, and the dial curve is a test-set analysis. **What is not closed.** The recipe leaves MVBench 3.6 and Video-MME 3.8 points below zero-shot; the KL anchor is LwF and the replay is rehearsal: the claim is the measurement, not a loss. We diagnose the localisation gap and do not close it; targets that penalise the video prior, longer context and onset training are future work. **Scope.** The negative results hold at 2B to 9B with our data (focal loss [13] untested); comparisons are among the policies and the 0.8B to 27B backbones we evaluate.

8. Conclusion

Streaming VAD models know the video and not the moment: every model from 0.8B to 27B separates anomalous videos at 96 AUC and ranks the moments inside them at 72, and the frame-level metric is mostly the first number. Fine-tuning strengthens the video prior, flattens the contrast and narrows the model while the metric rises; the narrowing is repairable, the contrast is not, by anything we tried. jev-hard, scored three ways and reported with retention and localisation, is how we propose the next method be judged.

LLM Usage Statement. An LLM helped draft text and code; the authors verified every number and claim.

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Knows the Video, Not the Moment: Temporal Localisation Is the Missing Capability of Streaming Video Anomaly Detection

Supplementary Material

9. Prompts

V1 (MoniTor), UCF-Crime. State: *"Surveillance camera footage, 2 seconds."* Question: *"If you were a law enforcement agency, would you rate this scene as a scene involving suspicious activities such as abuse (intentionally harming or mistreating others), arrest (legally detaining someone), arson (deliberately setting fire), assault (. . .), burglary (. . .), explosion (. . .), fighting (. . .), road accident (. . .), robbery (. . .), shooting (. . .), shoplifting (. . .), stealing (. . .), or vandalism (deliberate destruction of property), rather than a standard scene?"* Options: Yes, No. Score: $p(\text{Yes})$. The class definitions are MoniTor's [28].

V1 (MoniTor), XD-Violence. State: *"Video footage, 2 seconds."* Question: *"Would you rate this scene as a scene involving suspicious or potential criminal activities, such as abuse (inflicting improper harm or misconduct on others), car accident (collision events involving motor vehicles . . .), explosion (. . .), fighting (. . .), riot (a group of people protesting or resisting . . . in a violent manner), or shooting (act of firing with firearms . . .), rather than a normal scene?"* Options: Yes, No.

V2 (ESOM). Question: *"Which defined event happens in these 2 seconds? Answer with its option, or None if no defined event happens. An event counts from its first act (the first blow, flame, blast, entry or contact) until it stops."* followed by a table with one row per class, *Option | Counts | Does not count*, with anonymous letters A–M (UCF-Crime) or A–F (XD-Violence) and a last row *None: no defined event: people walking, talking, working, driving or standing as usual*. Options: the letters and None. Score: $1 - p(\text{None})$; the most probable letter names the class. The definitions follow ESOM [14]; None is placed last so that every letter maps to its own answer code on d3.

10. How the decision models were trained

The decision models were made in similar ways, as far as their reports and cards disclose. StartLux [36] trains all language-model parameters on 2.7 M decision records with a soft cross-entropy between target and predicted option distributions, options permuted, then adapters on 1.7 M more with the bidirectional mask on, because "changing the attention mask of a causally trained model can disrupt its predictions"; the vision tower is frozen and a temperature is fitted afterwards. A community reproduction on Qwen3.5-4B [17] trains a rank-64 LoRA on every language-model projection, including the DeltaNet projections, with a 256-code head and a cross-entropy with soft targets, on 867 k records. H2O-Lightning [5] ships a rank-64 LoRA. d3's weights, compared with their base, show every language-model matrix changed and the vision tower untouched. ReactVAU [2], the only one trained on anomaly data, trains a projector and a LoRA for one epoch with a binary cross-entropy on $\sigma(z_{\text{Yes}} - z_{\text{No}})$. A benchmark sweep [18] found that a rank-16 decoder LoRA transferred across image tasks (AI2D 0.652 to 0.707) while a vision-tower adapter helped little.

Model	Base	Params	Read-out	Prompt mask
d3-nano [23]	Qwen3.5-2B	2.2B	codes	noncausal
d3-flash [23]	Qwen3.5-9B	8.4B	codes	noncausal
d3 (27B) [23]	Qwen3.5-27B	27B	codes	noncausal
Clef-Flash [3]	Qwen3.5-9B	9.4B	schema head	causal
Qwen3.5-2B (control)	Qwen3.5-2B	2B	first token	causal
Qwen3.5-9B (control)	Qwen3.5-9B	9.7B	first token	causal

Table 5. **Models.** Decision-trained models of three sizes and one alternative read-out family, with the plain instruction models of the same backbone as controls. The zero-shot size curve of Sec. 6.3 adds d3-edge (0.8B), d3-lite (1.2B) and d3-mini (4B).

Model	Read-out	Zero-shot	Hard labels	mix+KL
Same size, one variable: decision post-training				
d3-nano (2.2B)	codes	82.7 / 41.9	87.6 / 35.6	87.9 / 47.4
Qwen3.5-2B (plain)	first token	72.0 / 35.7	87.6 / 33.5†	88.2 / 45.9
Scale				
d3-flash (8.4B)	codes	85.6 / 42.8†	89.2 / 38.4†	88.4 / 48.2
Qwen3.5-9B (plain)	first token	82.4 / 41.5†	89.5 / 44.4†	**89.1 / 49.1**
d3 (27B)	codes	84.7 / 41.8	–	88.0 / 46.6
Read-out family				
Clef-Flash (9B)	joint schema	84.1 / 36.5†	88.8 / 39.4†	88.3 / 49.0
Three seeds, mix+KL				
d3-nano	codes		88.4 ± 0.5 / 45.2 ± 2.1	
Qwen3.5-2B	first token		88.0 ± 0.3 / 44.3 ± 1.5	

Table 6. **The recipe holds across size, backbone type and read-out.** UCF-Crime 290, frame-level AUC / AP in %, one seed per cell unless stated; the same training windows and LoRA settings for every model. mix+KL matches or beats hard-label LoRA on AP everywhere, with the largest gain on the plain 2B backbone (+12.4), and the 27B model trained with it lands at the 2.2B level. †: deck pipeline, to be re-scored. –: not run.

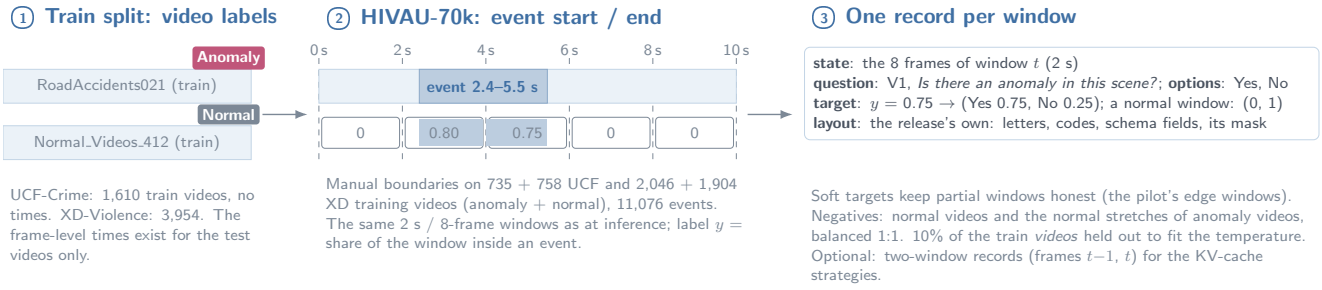


Figure 4. **From video labels to window records.** The UCF-Crime and XD-Violence training splits are labelled per video. HIVAU-70k [34] adds the start and end of every anomaly event on those videos. We cut the same 2 s windows of eight frames as at inference, label each by the share of the window inside an event, and write it in the model's own record format with that share as a soft target. The test videos are never read.

Online method	Trained on	UCF AUC	XD AP
ReactVAU [2]	UCF / XD, LoRA + SFT	88.4	88.5
Flashback [8]	–	87.3	75.1
REWARD [7]	UCF / XD, weakly sup.	86.9	77.7
ESOM [14]	–	86.2	71.7
MoniTor [28]	–	82.6	55.0
d3-flash (8.4B), zero-shot	–	85.6	81.4
d3-nano (2.2B), zero-shot	–	82.7	79.7
d3-nano, mix + KL	UCF	87.9	86.7‡
d3-nano, mix + KL + fov2	UCF	88.7	86.6‡
d3-nano, fovdial α=.25	UCF	89.0	86.6‡
d3-nano, mix + KL	XD	–	92.1
Qwen3.5-9B, mix + KL + fov2	UCF	**89.3**	–

Table 7. **Where this sits among online VAD methods.** Frame-level UCF-Crime AUC and XD-Violence AP in %; published methods as their papers report them, ours under the protocol of Sec. 3.1 on the single pipeline of Sec. 6.1. Offline methods read future frames and are not listed. ‡UCF-Crime adapter on XD-Violence without XD training.

11. Record formats

Every model receives the same frames and question text in its own record format. For Clef-Flash a record is {`state, images, questions: {answer: {type: choice, instructions: <question>, criteria: {Yes: null, No: null}}}}`; the release renders it as *system prompt* | *STATE:* | *image tokens* | *state* | *SCHEMA FIELDS* | *JOINT SCHEMA DECISIONS:* and the head returns one logit per option. For d3 the same fields go through its `system_one` call with the eight frames as images; the option probabilities are the softmax over the options' answer codes. StartLux and H2O-Lightning receive the question with lettered options and the probability is read from the letter (label) tokens at the last position; H2O-Lightning reads frames 0, 2, 5 and 7 of the window. The Qwen3.5-9B control receives a chat prompt ending in *"Answer with Yes or No."* and p(Yes) is the normalised next-token probability of the two option tokens.

12. Context across windows: an exact KV cache

Under the protocol a window is judged alone: the model has a KV cache within a window but no memory between windows. To test whether the missed events need the past rather than pixels, we compare six strategies that all score the same newest 2 s: **S0** the newest window only; **S1** recompute, the newest window plus the past L seconds re-read each time; **S2** sliding KV, the cache of the past L seconds kept and only the eight new frames prefilled (Fig. 5); **S3** sinks and sliding, as S2 with the first tokens of the stream kept [26]; **S4** compressed KV, as S2 with the past pruned to a fixed token budget [29]; **S5** text memory, the past windows' scores written into the state in words, as MoniTor does.

Implementation. The cache is implemented for the Qwen3.5 hybrid backbone in HuggingFace Transformers 5.17 with a `DynamicCache`. One cache is kept per video. At window 0 the record prefix (system prompt and *STATE:*) is prefilled. At every window the frames older than the policy's keep window are evicted from the full-attention layers by index selection on their keys and values, the sinks (the first s seconds) excepted; the gated DeltaNet layers' convolution and recurrent states are left as they are. The eight new frames are encoded by the processor and prefilled with explicit three-dimensional M-RoPE positions computed over the whole history of token ids, so that evicted frames keep their positions. For the compressed strategy (S4) the past frame segments, the newest excepted, are pruned pass by pass, oldest first, until the token budget is met. The question segment (*state* | *schema* | *suffix*) is encoded without

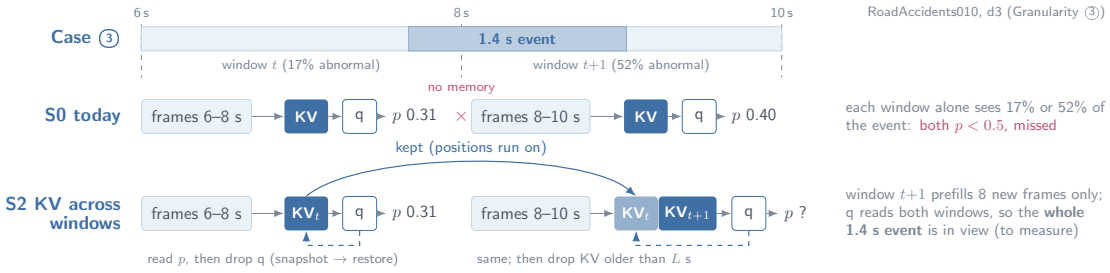


Figure 5. **A KV cache across windows, on a split event.** Top (S0): each window is its own request, so a 1.4 s event that covers 17% of one window and 52% of the next is scored twice below threshold. Bottom (S2): window t+1 prefills only its eight new frames on top of the cached frames of window t; the question is appended last and read, then its tokens are dropped again (snapshot → restore), so the cache holds frames only; positions run on and KV older than L seconds is evicted.

images, its spans shifted by the length of the cached history, appended after a deep copy of the cache, and the head is run on the hidden states of all cached tokens and the question's; the cache is then restored. On the first window of every video the layout is checked against the release's own encoding of the same record and the run aborts on a mismatch. On windows from which nothing has been evicted the probabilities match recomputation (S1) to within 1.7×10^{-3} over all 102 windows of the pilot.

13. Context across windows: the pilot table

Strategy	RA010	RA122	Shop037	Caught	P95 (ms)	RTF
S0 today	0.26 / 0.20	0.03 / 0.09	0.23 / 0.09	0 / 5	893	0.45
S1 recompute, L=2	0.20 / 0.30	0.03 / 0.22	0.32 / 0.35	0 / 5	1765	0.88
S1 recompute, L=4	0.12 / 0.19	0.03 / 0.32	0.33 / **0.40**	0 / 5	2388	**1.19**
S2 sliding KV, L=2	0.24 / **0.38**	0.02 / 0.24	0.19 / 0.20	0 / 5	984	0.49
S2 sliding KV, L=4	0.16 / 0.36	0.02 / 0.21	0.21 / 0.21	0 / 5	996	0.50
S3 sinks + sliding	0.15 / 0.21	0.02 / 0.12	0.21 / 0.21	0 / 5	1007	0.50
S4 compressed, 640 tok.	0.24 / 0.34	0.02 / 0.22	0.16 / 0.16	0 / 5	1011	0.51
S5 text memory	0.20 / 0.19	0.02 / 0.04	0.14 / 0.07	0 / 5	907	0.45

Table 8. **Context across windows on five split events (Clef-Flash, V1).** p(abnormal) of the two event windows (first / second) of three events; the other two stay ≤ 0.03 in every row. Caught: events with any window ≥ 0.5. No false positive on the 20 windows of five normal videos in any row. Latency on one GB10 with reference DeltaNet kernels; comparable within the table only.

14. Pilot videos

The ten UCF-Crime test videos of Tab. 8: RoadAccidents010 (event 7.67–9.03 s), RoadAccidents004, RoadAccidents011, RoadAccidents122 (10.00–12.00 s), Shoplifting037 (38.00–40.00 s), and the normal videos Normal_Videos_345, 881, 878, 745 and 889. The five events were chosen from the 33 events shorter than 4 s that cross a 2 s boundary as those that Clef-Flash misses with S0. RoadAccidents122 and Shoplifting037 end exactly on a window boundary, so their "second" window is labelled abnormal by one inclusive boundary frame while all eight sampled frames lie after the event; this is the label effect quantified in Sec. 6.3.

15. Per-class results on the jev-hard tiers

Small tier, UCF-Crime	windows	zero-shot	mix + KL	fov2	WiSE .5
Burglary	149	1.6	2.7	3.1	3.0
Shooting	103	1.8	2.7	3.1	4.4
Arrest	100	4.7	4.9	4.0	2.7
Shoplifting	93	1.4	2.3	2.4	1.6
Arson	82	0.5	0.8	1.0	1.5
Explosion	73	2.9	2.7	10.7	7.3
Stealing	65	20.2	46.2	72.8	81.8
Vandalism	26	2.5	4.6	5.0	3.3
Robbery	16	0.2	0.2	0.2	0.2
Fighting	13	9.6	19.8	15.5	5.1
RoadAccidents	8	0.4	0.1	0.1	0.2
Abuse	5	0.1	0.1	0.1	0.1

Table 9. **Small-tier AP per class (d3-nano).** Positives are the class's small-tier windows, negatives every negative window of the test set (chance 0.6%). Nine classes are at or near chance for every model; the tier score is carried by Stealing and Explosion.



Figure 6. **The coverage target.** A window's target is the fraction of its 2 s inside an annotated event, so edge windows are learnt as partial; the hard-label variant thresholds it at 0.5.

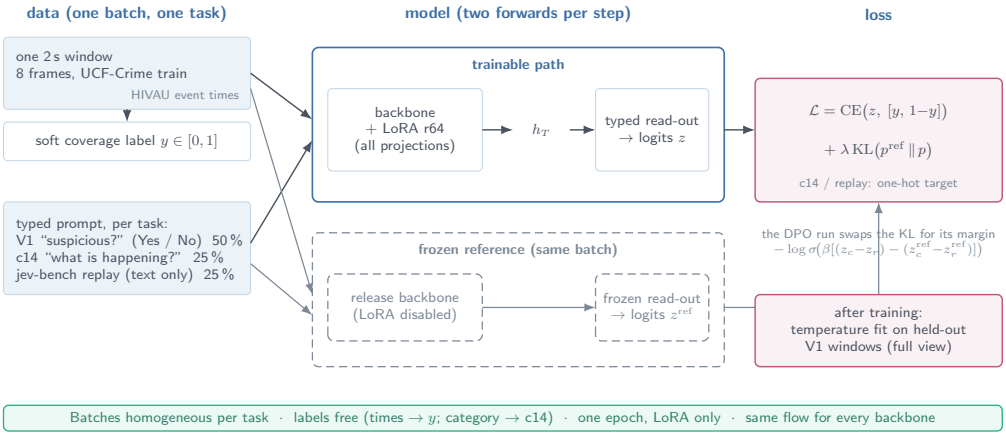


Figure 7. **One training step of mix + KL, with foveation.** The window's eight frames pass the conditional foveation of Sec. 5.3 (skipped when the moving area is large), join the record's typed question, and run twice: through the trainable LoRA path and through the frozen release. The loss is the soft cross-entropy on the model's own option read-out plus the KL to the release's probabilities. Without foveation the frames go in unchanged.

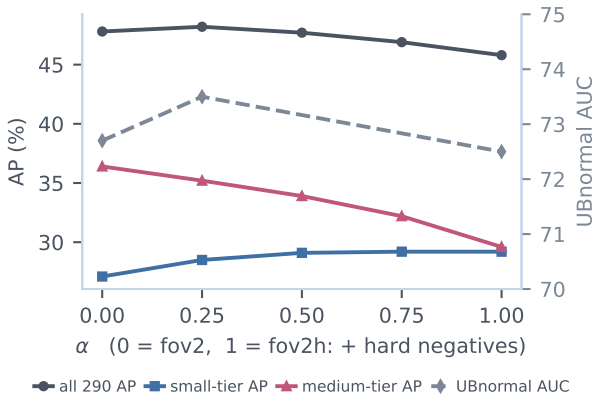


Figure 8. **The dial between fov2 and its hard-negative variant.** Interpolating the two adapters trades the medium tier for the small one along a smooth curve; the all-video AP peaks inside the dial and so does transfer to unseen scenes. One seed per point; the 0.4 AP between the peak and $\alpha = 0$ is inside seed noise.

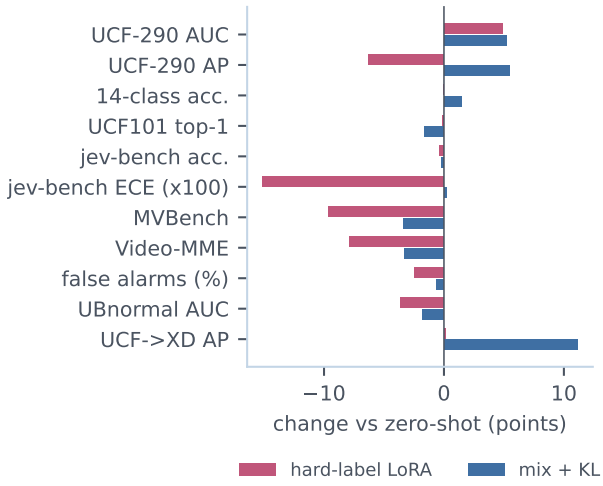


Figure 9. **What one fine-tune changes, on eleven measurements (d3-nano).** Change against zero-shot for the hard-label fine-tune and for the mix + KL recipe, in points, right is better (ECE and false alarms are sign-flipped). The hard-label fine-tune improves the task's AUC and worsens almost everything else; the recipe keeps the AUC, turns the AP and transfer around, and leaves MVBench, Video-MME and UBnormal partly open. The 14-class bar for the hard-label fine-tune is zero because that model never saw the question.

16. Unlabelled-window score by motion

17. The rest of the foveation ladder

18. Inference-time zoom: the negative result

Table 15 gives the full matrix behind the one line in Sec. 6.4. On the five short events of Sec. 14 a TGM crop with oracle timing lifts four of five over 0.5, so the mechanism is real; on the full test set the zoomed pass raises false positives as much as true positives, and the best policy ends 0.3 AUC and 1.2 AP below the model without zoom. With oracle timing and no crop the hard-label model catches 126 of 156 events (80.8%; zero-shot 64.1%): 30 events are missed with perfect timing.

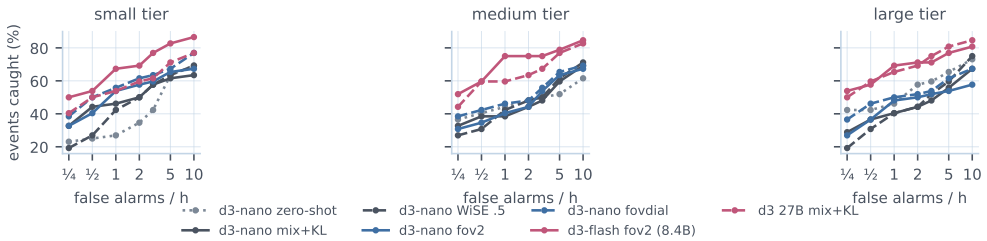


Figure 10. **Events caught against false alarms per hour, per tier (UCF-Crime).** The threshold is swept per model; an event is caught when any of its windows alarms. Size separates the curves at every operating point, the small tier is not the lowest, and the fine-tuned 2.2B models cross the zero-shot one only above two alarms per hour.

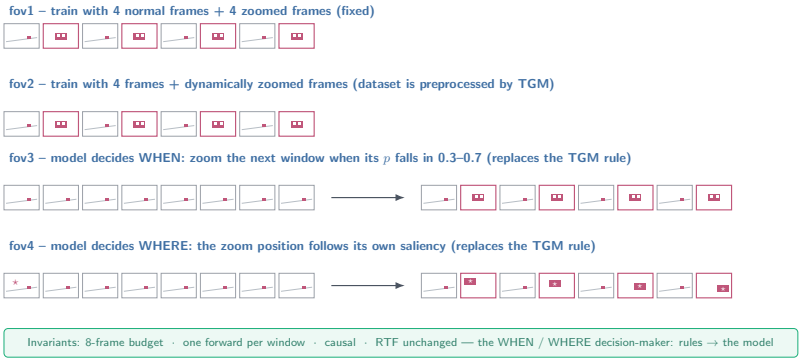


Figure 11. **The foveation ladder: what the model sees.** Grey, a full frame; red, a crop centred on the motion peak and up-scaled. fov1 always alternates full frames and crops. fov2 keeps the eight full frames when the moving area is at least 4.5% of the frame and alternates otherwise. fov3 lets the model's own uncertainty at window t trigger the crops at window $t{+}1$. fov4 takes the crop position from the model's gradient saliency instead of the motion peak. The frame count and the token count never change.

d3-nano on XD-Violence (800 videos)	small	medium	large	all
Trained on XD-Violence (20 k windows)				
hard labels	93.7 / 52.3	97.3 / 87.0	97.9 / 89.1	96.8 / 91.6
WiSE-FT α=.5 of it	94.5 / 58.1	97.7 / 88.6	98.1 / 90.0	97.2 / 92.6
mix + KL	94.4 / 61.3	97.5 / 87.2	97.8 / 88.6	96.9 / 92.1
mix + KL + fov2	94.5 / **62.0**	97.4 / 87.1	97.8 / 88.6	96.9 / 92.1
mix + KL, joint with UCF-Crime	93.9 / 58.3	97.4 / 87.0	97.8 / 88.6	96.8 / 91.8
Transferred from UCF-Crime, no XD training				
zero-shot	91.5 / 43.0	94.1 / 61.9	94.3 / 65.1	93.6 / 79.7
hard labels	91.9 / 45.1	89.7 / 54.9	90.3 / 59.8	90.4 / 75.7
WiSE-FT α=.5 of it	93.3 / 46.5	92.2 / 58.0	92.6 / 64.5	92.6 / 78.7
mix + KL	93.3 / 54.8	95.1 / 75.1	95.2 / 78.3	94.8 / 86.7
mix + KL + fov2	93.8 / 59.1	95.1 / 74.5	94.9 / 76.9	94.7 / 86.6
fovdial α=.25	94.1 / **61.5**	95.1 / 74.0	94.9 / 76.5	94.8 / 86.6

Table 10. **jev-hard on XD-Violence: frame AUC / AP per tier.** 1,238 events in terciles at 7.0% and 16.8% of the frame; 2,645 / 4,995 / 4,667 positive windows. Top: models fine-tuned on XD-Violence's training split. Bottom: the UCF-Crime adapters of Tab. 4 applied unchanged. One seed each.

d3-nano	windows	area Q1	Q2	Q3	Q4
mix + KL	anomalous videos, unlabelled	X	X	X	X
mix + KL	normal videos	X	X	X	X
fov2	anomalous videos, unlabelled	X	X	X	X
fov2	normal videos	X	X	X	X

Table 11. **Mean score by TGM hot-area quartile**, on 2,000 sampled UCF-Crime test windows. X: being computed.

Model	small	medium	large	all 290
d3-nano, zero-shot	76.1 / 23.1	87.7 / 29.5	89.5 / 21.7	82.7 / 41.9
d3-nano, hard labels	84.9 / **15.9**	89.0 / 19.0	91.0 / 16.3	87.6 / 35.6
d3-nano, mix + KL (3 seeds)	85.2 / 22.6	91.3 / 34.8	91.5 / 21.1	88.4 / 45.2
d3-nano, DPO	84.6 / 25.5	89.7 / 35.0	91.8 / 23.1	87.8 / 45.9
d3-nano, WiSE-FT α=.5	86.2 / 27.1	91.1 / 35.0	92.5 / 22.9	89.1 / 47.8
Qwen3.5-2B, mix + KL	85.4 / 25.5	89.7 / 30.1	91.9 / 33.7	88.2 / 46.4
d3-flash (8.4B), mix + KL	84.4 / 24.2	91.0 / 37.0	93.0 / 31.5	88.4 / 48.2
Qwen3.5-9B, mix + KL	85.4 / 27.3	92.3 / 38.8	92.4 / 24.6	89.1 / 49.5
Clef-Flash (9B), mix + KL	84.3 / 27.8	91.5 / 38.9	92.3 / 25.1	88.3 / 49.0
d3 (27B), zero-shot	78.8 / 21.5	88.4 / 23.0	91.7 / 25.0	84.7 / 41.8
d3 (27B), mix + KL	83.3 / 24.3	91.8 / 33.4	92.8 / 27.9	88.0 / 46.6

Table 12. **jev-hard on UCF-Crime: frame AUC / AP per tier.** 52 events per tier, every negative window kept. The small tier is the hardest for every model and no model exceeds 28 AP on it; 4.5× the parameters buy at most two points. The mix + KL base is the mean of three seeds (Tab. 4 gives the spread). The hard-label fine-tune trades the small tier (23.1 to 15.9) for the aggregate. Bold: the drop. X: not finished.

d3-nano	UBnormal	UCF→XD	XD→UCF	MVBench	Video-MME
zero-shot	70.7	79.7	82.7	57.3	63.9
hard labels	67.1	75.7	78.7	47.7	56.0
mix + KL (3 seeds)	68.9	**86.7**	**84.4**	53.7 ±0.7	60.1 ±0.6
mix + KL, UCF101 clips as replay	71.3	83.7	X	53.4	59.7
mix + KL, joint with XD	62.9	(91.8)	(88.3)	X	X
WiSE-FT α=.25	72.8	78.2	X	**56.0**	**63.0**
WiSE-FT α=.5	70.9	78.7	X	54.2	60.3
fov2 (3 seeds)	72.7 ±2.4	86.6	84.2	53.8	59.0
fovdial $\alpha = 0.25$	**73.5**	86.6	X	54.4	58.7

Table 13. **Generalisation matrix.** Frame AUC on UBnormal and on UCF-Crime with XD-trained adapters of the same recipe; frame AP on XD-Violence with the UCF adapters (the joint model has seen both, in parentheses); accuracy on MVBench and Video-MME short. Single measurements unless a spread is given. X: not run.

d3-nano, mix + KL base	small	medium	large	all 290
best inference-time zoom†	–	–	–	88.6 / 41.6
fov3, model gate	85.4 / 26.0	91.1 / 37.6	91.7 / 26.0	88.6 / 47.9
fov2, gate at 3% / 6% (inference), AP	27.4 / 27.1	36.9 / 36.8	24.4 / 24.3	48.2 / 47.9
fovdial $\alpha = 0.5$	86.2 / 29.1	90.9 / 33.9	91.8 / 23.1	88.9 / 47.7
fovdial $\alpha = 0.75$	86.1 / 29.2	90.4 / 32.2	91.6 / 21.9	88.7 / 46.9
fov4, saliency (103-video screen)	–	–	–	85.1 / 54.5
fov2, same 103 videos	–	–	–	86.2 / 56.0
Two forward passes per window (RTF 0.36)				
mean(base, fov2)	85.0 / 25.3	91.4 / 37.8	92.1 / 27.8	88.5 / 48.1
max(WiSE α=.5, fov2)	86.6 / 28.1	91.7 / 37.3	92.3 / 25.1	89.4 / 49.1

Table 14. **Foveation variants not in Tab. 4.** †: hard-label model with a centre crop on uncertain windows, earlier pipeline, where the un-zoomed model scored 88.9 / 42.8. fov4 was screened on a 103-video subset that keeps every small-tier event. fov3 fires on 7% of the windows, one window late.

	d3-nano zero-shot		d3-nano hard labels	
Policy	AUC	AP	AUC	AP
no zoom	82.8	39.8	**88.9**	**42.8**
TGM 1/3 crop, every window	82.1	37.9	84.9	36.8
TGM 1/3 crop, when $p < 0.2$	82.8	39.5	88.2	40.0
TGM 1/2 crop, every window	**83.8**	39.6	87.5	41.3
TGM 1/2 crop, when $p < 0.2$	82.9	**39.8**	88.6	40.9
centre 1/2 crop, when $p < 0.2$	–	–	88.6	41.6

Table 15. **Inference-time zoom on UCF-Crime (290 videos, earlier pipeline).** The crop is centred on the TGM peak (or the frame centre), a third or a half of the frame, up-scaled 2×, and the window is scored as the maximum of the un-zoomed and zoomed probabilities, on every window or only when the first pass is uncertain. No policy exceeds the un-zoomed model.